

Research paper

HBUED: An EEG dataset for emotion recognition

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ABSTRACT

Emotion recognition via electroencephalogram (EEG) data is crucial for improving human-computer interaction. In practice, researchers require a substantial quantity of EEG samples to train and validate models. However, existing EEG datasets typically have a limited number of subjects. To address this issue, we present a large-scale EEG dataset, the Hebei University Emotional EEG Dataset (HBUED), specifically designed for research on human emotion recognition. Furthermore, this research presents a deep learning methodology aimed at improving emotion recognition performance by efficiently handling complicated samples in EEG-based emotion recognition. This method first constructs a dual-input network architecture to extract discriminative features of EEG signals from two perspectives for classification. Furthermore, this paper uses a parallel feature extraction module for EEG signals, which increases the number of neurons per layer by expanding the network width, thereby extracting more comprehensive feature information while avoiding overfitting caused by excessive network depth. In addition, a topological feature extraction module has been created to better capture the topological characteristics of EEG signals. Lastly, the proposed method is validated on both the self-constructed HBUED and the public DEAP datasets, with experimental results demonstrating its effectiveness. The HBUED datasets and the source code of the proposed method are publicly available at: <https://tensorground.github.io/HBUED.github.io/>.

1. Introduction

With the rapid advancement of artificial intelligence, brain-computer interface (BCI) technology has garnered increasing interest. Emotion recognition, a key aspect of BCI technology, holds significant research value (Liu et al., 2023a, 2023b, 2023c, 2023d; An et al., 2023). Emotion recognition is categorized into two forms based on the signals: recognition through non-physiological signals and recognition through physiological signals. Non-physiological signals encompass vocal tone, body movements, and facial expressions. These signals are subject to subjective control and can be concealed, often resulting in classifier recognition errors (Zhao et al., 2022; Liu et al., 2021). Physiological signals include electrooculogram (EOG), electromyogram (EMG), and EEG. These signals are generated spontaneously and are difficult to hide or fake. Among these, EEG-based emotion recognition has garnered increasing academic interest owing to advancements in monitoring devices, resulting in notable success in recent years (An et al., 2023).

In the area of EEG-based emotion recognition, researchers often

quantify emotions using either discrete or dimensional models. The discrete emotion model posits that independent basic emotions or combinations of these basic emotions can represent all emotions. For instance, Ekman (1971) proposed a discrete emotion model comprising six fundamental emotions: happiness, surprise, sadness, fear, anger, and disgust. Building on this, Plutchik (2001) expanded the set of basic emotions to propose a discrete emotion model with eight basic emotions. However, because multiple factors influence the generation of emotions, it is widely believed that using a discrete, one-dimensional model to measure emotions is inadequate. To address this, Russell (2003) proposed a two-dimensional (2D) continuous emotion model known as the valence-arousal (VA) model, as seen in Fig. 1(a). In this model, the horizontal axis represents valence, which reflects the degree of pleasure associated with an emotion, with higher values indicating greater pleasure. For example, disappointment is an unpleasant emotion with low valence, whereas happiness is a pleasant emotion with high valence. The vertical axis represents arousal, measuring the level of activation related to an emotion. For example, sadness is classified as a

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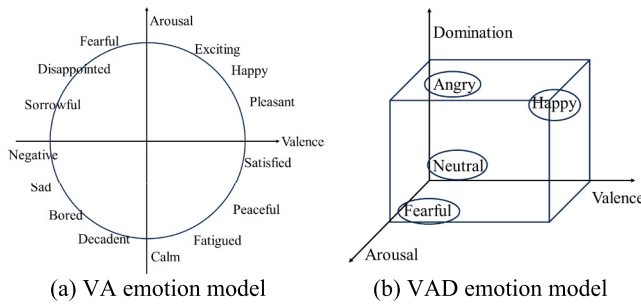


Fig. 1. Commonly used dimension-based sentiment models.

low arousal emotion, whereas fear is considered a high arousal emotion. Subsequently, [Mehrabian \(1996\)](#) extended the 2D continuous emotion model to three-dimensional (3D) continuous emotion model by adding a dominance dimension, resulting in the valence-arousal-dominance (VAD) emotion model, illustrated in [Fig. 1\(b\)](#). The dominance dimension reflects the extent of control or influence associated with an emotion, ranging from submissiveness (absence of control) to dominance (complete control). For example, when an individual experiences fear, they lack control over their surroundings and feel submissive. In contrast, when a person feels anger, they are in a dominant state, inclined to aggression, and exhibit a high level of dominance.

EEG datasets form the foundation for conducting EEG research. Currently, available EEG datasets include those focused on motor imagery, sleep detection, epilepsy, and emotional EEG data. Among these, publicly accessible emotional EEG datasets are relatively scarce and generally small in scale, with a limited number of participants. For instance, the DEAP dataset used Biosemi Active II equipment to collect EEG and other peripheral physiological signals ([Koelstra et al., 2011](#)). To create this dataset, we select 40 one-minute music videos to serve as emotional stimuli. After viewing each video segment, participants evaluated their emotional condition with the self-assessment manikin (SAM) questionnaire ([Bradley and Lang, 1994](#)). The DEAP dataset includes 32 participants, making it the largest publicly available EEG emotional dataset in terms of participant numbers. The MAHNOB HCI dataset provides EEG recordings and additional physiological signals from 27 subjects, with EEG data acquired using a 32-channel Biosemi Active II device ([Soleymani et al., 2011](#)). For this dataset, 20 emotive video clips sourced from films and online repositories are designed to elicit six emotions: disgust, amusement, joy, fear, sadness, and neutrality. The SEED dataset was collected using a 64-channel ESI NeuroScan system, selecting 15 Chinese movie clips as emotional stimuli to elicit three emotions: negative, neutral, and positive ([Zheng et al., 2017](#)). The SEED dataset includes 15 participants, with three emotion induction experiments conducted on each participant on three different days, resulting in multiple EEG recordings.

Researchers have carried out many emotion recognition studies using publicly available EEG datasets, which have advanced BCI technology. However, the existing EEG datasets used for emotion recognition typically involve a small number of participants and fail to capture the variations in individual EEG data adequately. As a result, when the model encounters individuals with specific characteristics, its performance will not be satisfactory. To overcome this limitation and create a large-scale EEG dataset that captures significant individual variability for human emotion recognition research, we detail the design and collection of the HBUED EEG dataset. Furthermore, a deep learning approach specifically designed for complex samples in EEG emotion recognition is introduced. This paper's principal contributions are delineated as follows:

- (1) Large-Scale EEG Dataset: We compiled a comprehensive EEG dataset for the study of human emotion recognition, comprising 50 participants (24 men and 26 females).

- (2) Dual-Input Deep Learning Model: We proposed a dual-input deep learning framework for EEG-based emotion recognition. The constructed parallel feature extraction module (PFEM) addresses the issue of overfitting that can arise from excessively increasing network depth by instead expanding network width. The topological feature extraction module (TFEM) efficiently utilizes the topological characteristics of EEG signals.
- (3) Extensive Experimental Validation: We performed comprehensive experiments utilizing both our proprietary dataset and publicly accessible datasets. The experimental results demonstrate the constructed dataset's validity and the proposed method's effectiveness.

This paper is structured as follows: [Section 2](#) reviews related works on EEG-based emotion recognition. [Section 3](#) delineates the data acquired process for the HBUED EEG dataset. [Section 4](#) gives detailed information on the structure of the proposed method. [Section 5](#) presents the experimental setup, gives the experimental results of the proposed method's performance on the HBUED and DEAP datasets, and shows the practicability of the constructed dataset. Finally, [Section 6](#) offers a succinct overview of the study and examines possible avenues for further research.

2. Related work

In recent years, EEG-based emotion recognition has experienced rapid advancements. Early research primarily relied on traditional classifiers for emotion recognition, such as support vector machines (SVM), K-nearest neighbors (KNN), decision trees (DT), and random forests (RF). For instance, [Wang et al. \(2011\)](#) transformed EEG signals into frequency domain features and used SVM, KNN, and multi-layer perceptron (MLP) for emotion classification, achieving favorable results. However, traditional methods for emotion recognition are inadequate in extracting deep features from EEG data, which limits the further enhancement of the performance of the emotion recognition model.

With the demonstrated success of deep learning across various fields, it has also been applied to EEG-based emotion recognition. For instance, [Li et al. \(2016\)](#) integrated convolutional neural networks (CNNs) with recurrent neural networks (RNNs) to construct a multi-channel EEG emotion recognition model. This model utilizes wavelet and scale transforms to encapsulate multi-channel EEG signals into grid-like frames, and then leverages the contextual information within these frames for feature extraction, thereby improving emotion recognition accuracy. [Liu et al. \(2020\)](#) introduced a dynamic differential entropy (DE) technique that integrates the benefits of empirical mode decomposition and DE for extracting EEG features, which were then classified using a CNN, resulting in favorable outcomes. [Guo et al. \(2023\)](#) extracted DE features from raw EEG data and employed phase synchronization indices to derive functional connectivity matrices, which were subsequently input into a temporal feature grouping attention model for classification, therefore improving emotion recognition accuracy. [Zhang and Cheng \(2024\)](#) presented an emotion recognition model utilizing CNN and Transformer architectures. This model designed a spectrum sum (SS) feature, requiring only the SS and DE features from 14 EEG channels to be input into the network. The CNN was used to learn local features, while the Transformer encoder learned global features, resulting in effective emotion recognition. [Chao and Dong \(2020\)](#) proposed an EEG emotion recognition method based on EEG feature rearrangement. This method initially transforms the information from all channels into a 3D feature matrix based on the locations of the electrode sensors. It then employed an advanced CNN, featuring both univariate and multivariate convolution layers, to extract features from the 3D matrix for emotion classification, resulting in favorable outcomes.

Although these CNN-based emotion recognition algorithms have shown promising results, CNNs have inherent limitations, particularly in fully capturing the relationships between EEG channels. Using graph

convolutional networks (GCNs) for EEG signal processing has gained momentum to extract topological information from EEG signals better. GCN may concurrently integrate both local and global topological information in EEG data, offering an exhaustive description of the relationships between local and global brain areas. As a result, more researchers are investigating graph convolutional architectures for emotion recognition. For example, Song et al. (2018) devised a dynamic graph convolutional neural network (DGCNN) to represent the interconnections among channels. This approach adaptively discerns the inherent relationships across EEG channels using a modifiable adjacency matrix, resulting in effective emotion recognition. An et al. (2024) presented a cross-subject emotion recognition technique by utilizing dynamic graph convolution and multi-source selective dynamic domain adaptation. This method adaptively identifies the inherent connections among various EEG channels through the adjacency matrix, mitigates transfer noise, and accelerates network performance, enhancing the emotional differentiation of domain-invariant features and augmenting the precision of cross-subject EEG emotion recognition. Yin et al. (2020) proposed a novel deep emotion recognition model combining GCN and Long-short memory networks (LSTMs). This model utilizes several GCNs to extract graph domain properties from EEG data, then employs LSTM to capture temporal features, enhancing emotion recognition. Gu et al. (2023) utilized generative adversarial networks (GAN) to generate latent representations of EEG signals and constructed dynamic adjacency matrices to illustrate the inherent links among brain channels. They utilized GCN and LSTM to identify emotions from EEG signals, enhancing the model's generalization capability. Due to emotion recognition models based solely on GCN tend to encounter overfitting issues (Choi et al., 2025), this paper combines the strengths of GCN and CNN, proposing a hybrid network based on CNN and DGCNN for EEG emotion classification.

3. Dataset construction

3.1. Subject information

Research has shown that participants with an extroverted and stable personality type are more likely to experience accurate emotional responses during emotion experiments (Zheng and Lu, 2015). Therefore, the Eysenck Personality Questionnaire (EPQ) was used to select suitable participants for this study (Eysenck et al., 1985). At the study's outset, 70 undergraduate and graduate students from the School of Electronic and Information Engineering at Hebei University were initially recruited for participant screening. All participants volunteered for the study and were fully informed of the experimental procedures and objectives. During the screening phase, the EPQ was administered to assess personality traits across four dimensions: extraversion (E), neuroticism (N), psychoticism (P), and lie scale (L). Statistical analysis of the questionnaire results revealed that 45 individuals exhibited an extroverted-stable personality type (i.e., high scores in extraversion, low scores in neuroticism, and normal-range scores in psychoticism). In addition, 10 participants were identified as extroverted-unstable, five as introverted-stable, while the remaining 10 were preliminarily excluded due to excessively high scores on the lie scale or pronounced psychoticism tendencies. Given the specific personality profile requirements of the emotion experiment, priority was given to selecting participants from the 45 extroverted-stable individuals. To ensure gender balance and diversity within the sample, an additional 5 participants with introverted-stable traits were included. The final cohort comprised 24 males and 26 females, achieving an approximately 1:1 gender ratio to minimize the potential confounding effect of gender on the experimental outcomes.

All 50 participants ultimately selected for the study met the following inclusion criteria: normal or corrected-to-normal vision, and no history of the following conditions as determined through a preliminary medical screening: (1) psychiatric disorders such as



Fig. 2. Experimental environment.

schizophrenia, bipolar affective disorder, depression, borderline personality disorder, or other personality and mental health disorders; (2) neurological diseases, including but not limited to traumatic brain injury, cortical ischemia, carotid artery disease, vascular headaches, cortical paralysis, epilepsy, brain tumors, dementia, chronic meningitis, multiple sclerosis, pernicious anemia, normotensive hydrocephalus, Parkinson's disease, and Huntington's disease. The participants ranged in age from 19 to 29 years, with a mean age of 22.28 years, spanning from lower-year undergraduates to postgraduate students. This age range effectively captures a representative cross-section of the young adult population. All participants took part in the experiment voluntarily and received compensation upon completion. The entire experimental process adhered strictly to ethical guidelines and was approved by the Ethics Committee of the Affiliated Hospital of Hebei University. Fig. 2 presents photographs taken of the participants in the experimental setting prior to the start of the procedure.

3.2. Stimulation information

Emotion induction plays a critical role in emotional perception experiments, with standard methods including static image stimuli, auditory stimuli, and video stimuli. For instance, the DEAP dataset employs music clips for emotional stimulation, while the SEED dataset utilizes video clips. The latter is considered an ideal method for eliciting emotional responses, as it integrates both visual and auditory modalities, thereby activating emotional reactions more comprehensively. Based on psychological theories and prior research, the selection of video stimuli in this study adhered to the following criteria:

First, regarding theoretical rationale and content criteria, the selection of video stimuli was based on Russell's Circumplex Model of Affect, constructing the stimulation system along the two dimensions of arousal and valence. High arousal videos included strong sensory stimuli designed to activate the sympathetic nervous system and elevate heart rate. In contrast, low arousal videos employed calm and static elements to induce a parasympathetically dominant state of relaxation. In terms of valence, high valence content was selected to convey positive emotions, aligning with Fredrickson's Broaden-and-Build Theory of Positive Emotions (Fredrickson, 2001). Conversely, low valence videos featured scenes of interpersonal conflict intended to elicit sadness or anger, in accordance with Lazarus's Cognitive Appraisal Theory of Emotion (Lazarus and Smith, 1988). Additionally, the selected stimuli prioritized themes drawn from popular culture familiar to college students, thereby minimizing cultural barriers that could compromise the effectiveness of emotion induction. Content involving extreme political or religious themes was excluded to reduce interference from individual differences.

Second, regarding the stimulus modality and parameter standards,

Table 1
Detailed information on video stimuli used for dataset construction.

Target emotion category	Duration	Frame rate	Video content description
High Valence High Arousal	120 s	15 fps	Olympic hurdling gold medal clip
			Olympic shooting gold medal clip
			Scene from <i>City of Rock</i>
			Scene from <i>Lost on Journey</i>
			Scene from <i>The Mermaid</i>
High Valence Low Arousal	120 s	15 fps	Women's football match clip
			Grassland scenery video
			Seaside travel video
			Waterfall scenery video
			Afternoon relaxation video
Low Valence High Arousal	120 s	15 fps	Violin music video
			Calming nature landscape video
			Scene from <i>Cry Me a Sad River</i>
			Scene from <i>Fist of Fury</i>
			Scene from a wartime documentary
Low Valence Low Arousal	120 s	15 fps	Scene from <i>The Ring</i>
			Scene from <i>The Bravest</i>
			Scene from a documentary on Japan's Unit 731
			Closing credits clip from a film
			Scene from a documentary on Miao silver crafting
			Micro-video of classroom teaching
			Micro-video on classical poetry interpretation
			Micro-video of sheepskin drum dance
			Micro-video of spinning marbles

this study drew on the design principles of the SEED dataset, which demonstrated that video stimuli—by integrating both visual and auditory information—are more effective in eliciting emotional responses. All video clips used in the experiment were fixed at 2 min, a length supported by Kalamangalam et al., as it is sufficient to reliably induce emotional states without causing attentional fatigue due to excessive length. The frame rate was standardized at 15 frames per second to ensure smooth playback while reducing visual load. In addition, the display resolution was deliberately lowered to 640 × 480 pixels to mitigate eye strain, thereby improving the quality of EEG data. The audio volume was normalized to 60 dB across all video stimuli to maintain consistency in auditory stimulation intensity.

Furthermore, a dual-stage validation mechanism involving experts and participants was employed in terms of the selection process and quality control. In the first stage, three clinical psychologists evaluated 100 candidate video clips using a scoring rubric (ranging from 1 to 5) based on arousal level, valence, and content appropriateness for emotional elicitation. Based on these scores, 30 high-potential segments were initially selected. A pilot study was conducted in the second stage with 10 informal participants from the same target population. Emotional elicitation effectiveness was assessed using both self-reported ratings on the self-assessment manikin (SAM) scale and physiological responses, including heart rate and skin conductance. Ultimately, 24 clips—representing a 24 % selection rate—were confirmed to be effective. These included six videos for each emotional category, ensuring balanced coverage.

Lastly, in terms of ethical and safety considerations, all low valence videos underwent strict content review to eliminate extremely violent imagery that could potentially cause psychological distress. An emergency stop mechanism was implemented during the experiment, allowing participants to terminate viewing negative emotional stimuli at any time. All video materials were sourced from public copyright libraries or obtained through proper authorization, ensuring full compliance with academic research copyright regulations.

Psychological experts meticulously selected the emotional stimuli used in this study based on the aforementioned criteria and consisted of 24 video clips, categorized into four groups: six videos targeting high

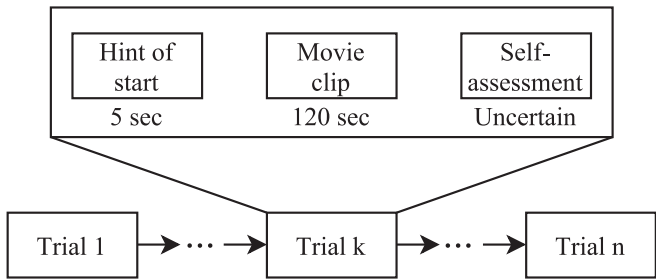


Fig. 3. Experimental flow.

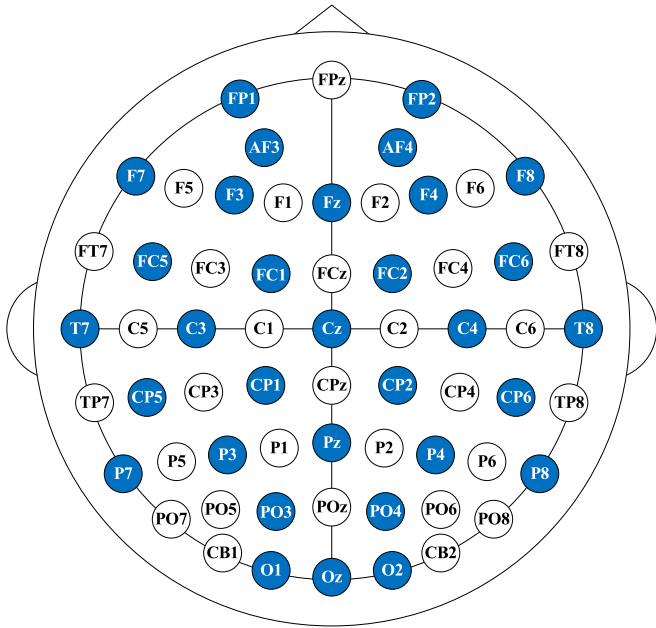


Fig. 4. Electrode positioning for the international 10–20 system (blue denotes the location of the 32 electrodes in this study). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

arousal and high valence, six targeting high arousal and low valence, six targeting low arousal and high valence, and six targeting low arousal and low valence. Each clip lasts for two minutes. During the experiment, measures such as reducing the display resolution were implemented to optimize the viewing experience and minimize external interference. Detailed information on the video stimuli is presented in Table 1.

3.3. Experimental procedure

Participants were instructed to get sufficient sleep the day before the experiment. Before the trial commenced, participants were required to sign an informed permission form and complete a questionnaire that gathered personal information, including any history of neurological or psychiatric disorders, age, and gender. Since emotion labels were self-assessed by the participants, they needed to understand and learn the basic self-assessment method before the experiment. Participants were requested to assess their emotional state on a scale from 1 to 9 across two dimensions: valence and arousal. Additionally, participants were briefed on the overall experimental procedure. Finally, they were asked to find a comfortable position and remain as still as possible throughout the experiment. The specific experimental procedure is shown in Fig. 3.

The experiment was divided into four parts, with participants allowed to rest between each part or end the break early if they wished. Each part consisted of six trials, and each trial included a 5-second start

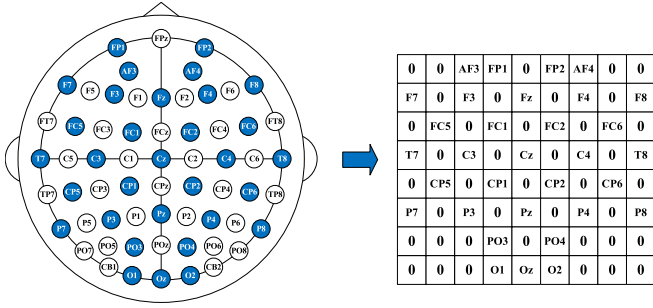


Fig. 5. Positional distribution of the spatial matrix.

signal, followed by a two minute video clip. After the video, the interface automatically switched to the self-assessment screen, where participants could take as much time as needed. EEG signals were continuously captured during the experiment by utilizing a 32-channel EEG equipment (Brain Product, BP) at a sampling rate of 1000 Hz, maintaining impedance below 5k Ω . Electrodes were strategically placed at 32 locations according to the worldwide 10–20 system, encompassing the left and right temporal, frontal, parietal, and occipital lobes. The specific electrode placement is shown in Fig. 4.

4. Method

4.1. Pre-processing

The acquired EEG data underwent preprocessing by using the EEGLAB toolbox. The preprocessing procedures included using a notch filter with a bandstop of 49–51 Hz to eradicate 50 Hz power line interference, succeeded by bandpass filtering within a range of 0.1–48 Hz to refine the data and eliminate artifacts. These data were subsequently downsampled to 128 Hz and segmented. The three seconds preceding the experiment were used as baseline signals for baseline correction, reducing biases caused by signal drift. Independent component analysis (ICA) is a statistical method employed to segregate multi-dimensional signals, aiming to decompose mixed signals into independent components. This work utilized ICA approaches to remove noise components,

including eye movements and prominent eye blinks. A finite impulse response (FIR) filter was employed to categorize the EEG signals into four frequency bands: θ (4–8 Hz), α (8–14 Hz), β (14–31 Hz), and γ (31–45 Hz).

4.2. Feature mapping

The self-collected and public datasets employed a 1-second sliding window to extract frequency domain information. In EEG signal processing, DE is frequently utilized as a principal method for feature extraction, as it represents the probability distribution of continuous signals across time (Liu et al., 2023a, 2023b, 2023c, 2023d). The DE feature can be articulated as:

$$H(x) = - \int_x f(x) \log(f(x)) dx \quad (1)$$

where x denotes the random variable, and $f(x)$ signify its probability density function. When x follows a Gaussian distribution $N(\mu, \sigma^2)$, the DE feature can be expressed as follows:

$$H(x) = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\delta^2}} \exp\left(-\frac{(x-\mu)^2}{2\delta^2}\right) \log \frac{1}{\sqrt{2\pi\delta^2}} \exp\left(-\frac{(x-\mu)^2}{2\delta^2}\right) dx \\ = \frac{1}{2} \log 2\pi e \delta \quad (2)$$

where μ and δ denote the expectation and variance of x , respectively.

Next, we perform Z-score normalization on $H(x)$, that is

$$Z = \frac{H(x) - \mu_h}{\delta_h} \quad (3)$$

where μ_h and δ_h denote the expectation and standard deviation of $H(x)$, respectively. The normalized DE feature is used as one of the network's inputs. Additionally, to provide the network with information from multiple perspectives, this input is further processed to generate another input for the network. The DE features are specifically represented in a 2D spatial matrix according to the electrode distribution of the EEG device, as illustrated in Fig. 5.

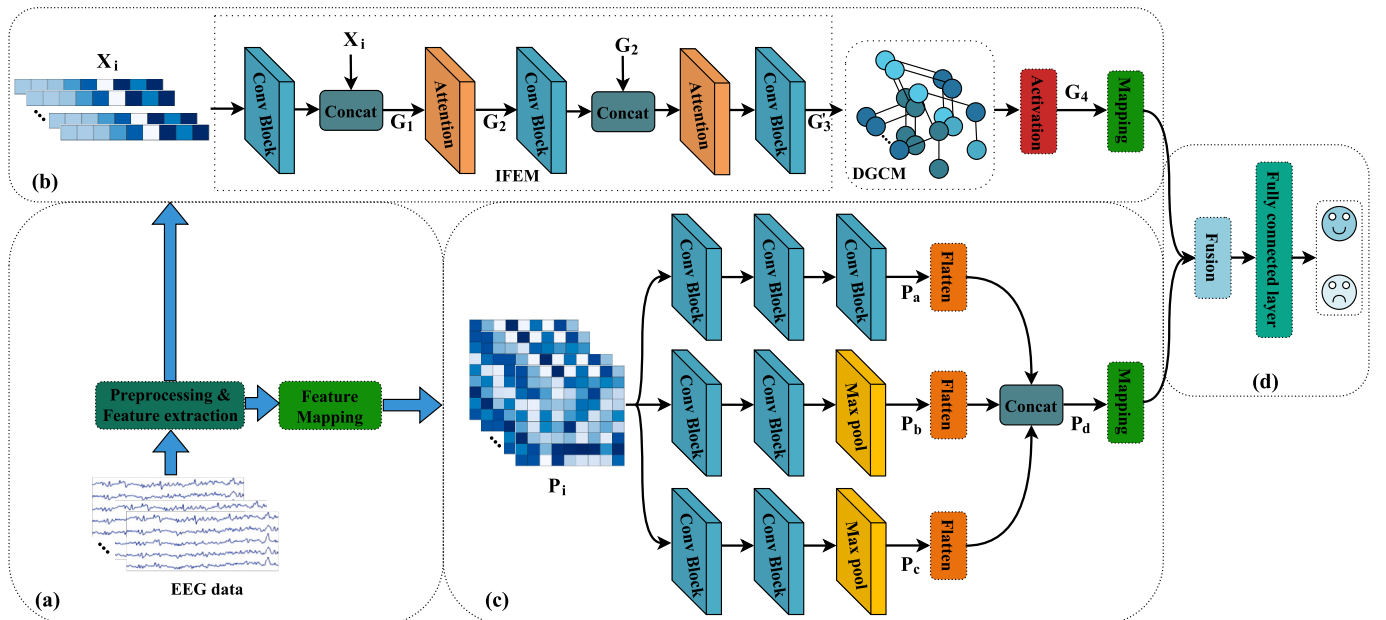


Fig. 6. The network architecture of the proposed method. (a) represents the operation process of preprocessing and feature extraction, (b) is the topological feature extraction module, (c) is the parallel feature extraction module, and (d) is the final fusion and sentiment classification.

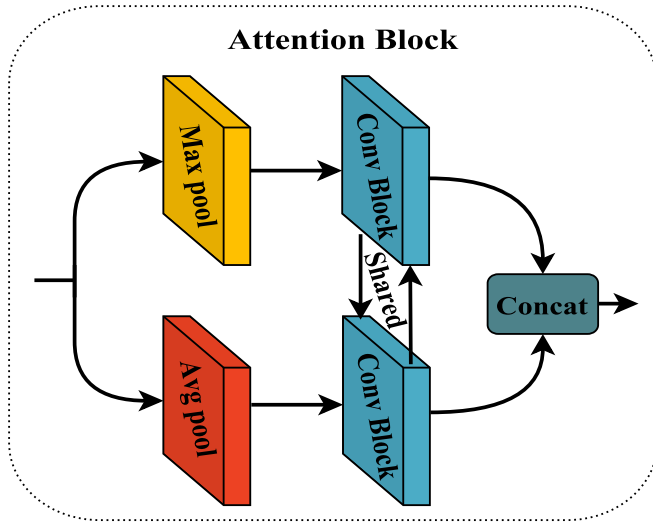


Fig. 7. Structure of the attention mechanism.

4.3. Feature extraction and classification

The classification model constructed in this study is a dual-input structure. The first input to the network consists of the signals with extracted DE features, while the second input is the DE features mapped onto a spatial matrix. The first input undergoes processing using a TFEM to derive topological information from the EEG signals, whilst the second input is directed through a PFEM to acquire deep spatial information from the EEG signals. The information from these two perspectives is subsequently integrated into a Softmax classifier for the final emotion classification. Fig. 6 illustrates the detailed structure of the proposed model. In the following sections, we will briefly introduce the TFEM, the PFEM, and the classifier.

4.3.1. Topological feature extraction module

In this study, we proposed the TFEM to extract EEG features and comprehensively identify key emotion-related channels. The constructed module comprises two components: an initial feature extraction module (IFEM) and a dynamic graph convolution module (DGCM). The IFEM primarily uses the attention mechanism to extract essential channel information relevant to emotions, thus preparing the input for the DGCM, which further explores inter-channel correlations. The main processing steps of the TFEM are outlined as follows.

In this study, for the input DE feature $\mathbf{X}_i \in \mathbb{R}^{C \times F}$, $i = 1, 2, \dots, T$, an initial feature extraction is first performed by using an IFEM based on an attention mechanism. The IFEM mainly comprises a convolutional block and an attention computation block, as shown in Fig. 6. In Fig. 6, the input data first undergoes convolution to produce output $\mathbf{G}_1$. To recover shallow features that might be lost during convolution, the output is concatenated with the input to generate $\mathbf{G}_1$. This is then passed through the attention layer to produce the final output feature $\mathbf{G}_2$. The structure of the attention mechanism is shown in Fig. 7.

The entire computational process of the IFEM can be summarized as follows:

$$\begin{cases} \mathbf{G}_1 = \sigma(\text{Bn}(\text{Conv1}(\mathbf{X}_i))) \\ \mathbf{G}_1 = \mathbf{X}_i \parallel \mathbf{G}_1 \\ \mathbf{G}_2 = \sigma(\text{Conv2}(\text{Maxpool}(\mathbf{G}_1)) \parallel \text{Conv2}(\text{Avgpool}(\mathbf{G}_1))) \end{cases} \quad (4)$$

where $\text{Conv1}()$ and $\text{Conv2}()$ represent different convolution operations, $\text{Bn}()$ denotes the batch normalization process, $\text{maxpool}()$ and $\text{avgpool}()$ denotes max pooling and average pooling respectively, $\sigma()$ represents the activation function, and $\parallel$ indicates the concatenation operation. To

fully extract EEG features, the aforementioned structure is stacked into two layers, resulting in the final feature output $\mathbf{G}_3$. Subsequently, feature selection is performed on $\mathbf{G}_3$ through a combination of convolution and batch normalization, as follows:

$$\mathbf{G}_3' = \sigma(\text{Bn}(\text{Conv}(\mathbf{G}_3))) \quad (5)$$

where $\text{Conv}()$ is convolution operation, $\text{Bn}()$ denotes the batch normalization process, $\sigma()$ represents the activation function.

To fully explore the connectivity relationships among EEG channels, the extracted features are fed into a DGCM for deep feature extraction. Dynamic graph convolution allows for a more flexible capture of the connectivity between EEG channels, effectively overcoming the limitations of manually setting the adjacency matrix in GCN. The computational procedure can be articulated as follows:

$$\mathbf{G}_4 = g(\mathbf{L}')\mathbf{G}_3' = \mathbf{U}g(\mathbf{A}')\mathbf{U}^T\mathbf{G}_3' \quad (6)$$

where $\mathbf{L}' = \mathbf{D} - \mathbf{W}'$ denotes the Laplacian matrix, $\mathbf{D}$ represents the degree matrix, $\mathbf{W}'$ is the adjacency matrix, and $\mathbf{A}' = \text{diag}([\lambda_0, \lambda_1, \dots, \lambda_{C-1}])$ represents the eigenvalues of matrix $\mathbf{L}'$, while $\mathbf{U}$ is the eigenvectors associated with these eigenvalues. To diminish the computing complexity of $g(\mathbf{A}')$, a K -order Chebyshev polynomial is utilized for approximation, rendering the approximate computation of $g(\mathbf{A}')$ defined as follows (Defferrard et al., 2016):

$$g(\mathbf{A}') = \sum_{i=0}^{K-1} \theta_i T_i(\tilde{\mathbf{A}}') \quad (7)$$

where $\tilde{\mathbf{A}}' = 2\mathbf{A}'/\lambda_{\max} - \mathbf{I}_C$ represents the normalized Laplacian matrix, $\lambda_{\max}$ is the maximum value in $\mathbf{A}'$, $\mathbf{I}_C$ is the identity matrix, and θ_i denotes the coefficients of the Chebyshev polynomials. The recursive formula for $T_i(x)$ is as follows:

$$\begin{cases} T_0(x) = 1, T_1(x) = x \\ T_i(x) = 2xT_{i-1}(x) - T_{i-2}(x), i \geq 2 \end{cases} \quad (8)$$

Thus, the final computation process of the DGCM can be defined as follows:

$$\mathbf{G}_4 = g(\mathbf{L}')\mathbf{G}_3' = \mathbf{U}g(\mathbf{A}')\mathbf{U}^T\mathbf{G}_3' = \sum_{i=0}^{K-1} \theta_i T_k(\tilde{\mathbf{L}}')\mathbf{G}_3' \quad (9)$$

where $\tilde{\mathbf{L}}' = 2\mathbf{L}'/\lambda_{\max} - \mathbf{I}_C$, θ_i denotes the coefficients of the Chebyshev polynomials. By configuring the adjacency matrix as an updatable parameter during network training, a more accurate representation of the relationships among EEG channels is achieved. The specific steps for updating the adjacency matrix will be discussed in detail in Section 4.3.3.

4.3.2. Parallel feature extraction module

The role of the PFEM is to extract deep spatial information from the spatial matrix for emotion classification. Generally, deeper networks perform better within a certain range, allowing for layer-by-layer abstraction and refinement of extracted knowledge. Considering the substantial number of subjects and the complexity of the information in the HBUED dataset, it is essential to increase the number of neurons and the amount of extracted information. However, in the field of EEG signal processing, deeper networks may lead to overfitting (Chen et al., 2022; Chen et al., 2023).

To address this issue, we constructed a PFEM, which is shown in Fig. 6. Moreover, most existing methods adopt generic feature extraction strategies that fail to fully exploit the unique characteristics of EEG data. In contrast, the proposed PFEM module is specifically designed to accommodate the 8×9 matrix form of DE-mapped EEG features. It employs multiple branches with different convolution kernel combinations to extract features from varying receptive fields, thereby

enhancing the ability to capture complex patterns in EEG signals and ensuring comprehensive coverage of emotion-related features. These features are then integrated through a concatenation process following the distinct branch structures. In many existing parallel feature extraction methods, lacking targeted branch design often leads to redundant or incomplete feature representations during fusion. Each branch is tailored to extract distinct aspects of EEG features in this work. This enables the fused representation to contain complementary information and more holistically reflect the multidimensional emotional characteristics embedded in EEG data. This, in turn, provides a more discriminative feature representation for downstream emotion recognition tasks. Furthermore, through the strategic configuration of convolutional kernels, batch normalization, and activation functions, the proposed module effectively suppresses noise while adaptively learning the dynamic patterns of EEG signals under different emotional states. This enhances the model's ability to cope with the complex distribution of EEG data—an aspect often overlooked in parallel feature extraction approaches.

The PFEM consists of three branches, each constructed using different convolution and pooling layer combinations. Specifically, the first branch comprises three convolutional blocks, the second branch includes two convolutional blocks followed by a max-pooling layer, and the third branch consists of one convolutional block and one max-pooling layer. Initially, the input DE features are mapped into a two-dimensional spatial structure as described in Section 3.2, resulting in a 3D representation denoted as $\mathbf{P}_i \in \mathbb{R}^{8 \times 9 \times F}, i = 1, 2, \dots, T$. This feature map is then fed into the PFEM, with the detailed computation process outlined as follows:

$$\begin{cases} \mathbf{P}_a = \sigma(\text{Bn}(\text{Conv}_{p3}(\mathbf{P}_1))) \\ \mathbf{P}'_1 = \sigma(\text{Bn}(\text{Conv}_{p2}(\mathbf{P}_1))) \\ \mathbf{P}''_1 = \sigma(\text{Bn}(\text{Conv}_{p1}(\mathbf{P}_1))) \end{cases} \quad (10)$$

$$\begin{cases} \mathbf{P}_b = \sigma(\text{Bn}(\text{Conv}_{p5}(\mathbf{P}_2))) \\ \mathbf{P}'_2 = \sigma(\text{Bn}(\text{Conv}_{p4}(\mathbf{P}_2))) \\ \mathbf{P}''_2 = \text{Maxpool}(\mathbf{P}_2) \end{cases} \quad (11)$$

$$\begin{cases} \mathbf{P}_c = \sigma(\text{Bn}(\text{Conv}_{p7}(\mathbf{P}_3))) \\ \mathbf{P}'_3 = \sigma(\text{Bn}(\text{Conv}_{p6}(\mathbf{P}_3))) \\ \mathbf{P}''_3 = \text{Maxpool}(\mathbf{P}_3) \end{cases} \quad (12)$$

where $\text{Conv}_{pi}()$, $i = 1, 2, \dots, 7$ denotes the convolution operation, while $\mathbf{P}_a, \mathbf{P}_b$ and $\mathbf{P}_c$ represent the output features from the three respective branches. $\text{Bn}()$ denotes the batch normalization process, $\sigma()$ represents the activation function. All output features are then flattened and concatenated to form the combined feature representation $\mathbf{P}_d$, which serves as a more comprehensive characterization of the input data. Moreover, the multi-branch parallel structure increases the network's width, ensuring that the fused features capture sufficient information for emotion classification. The incorporation of max-pooling not only facilitates the extraction of effective emotional information from EEG signals but also reduces the high dimensionality of EEG data, thereby mitigating the risk of overfitting.

Subsequently, the output features obtained in this section are concatenated with those from the TFEM to form the final emotional feature representation:

$$\mathbf{O} = \text{Mapping1}(\mathbf{G}_4) \parallel \text{Mapping2}(\mathbf{P}_d) \quad (13)$$

where $\mathbf{O}$ denotes the final concatenated emotional feature representation, while $\text{Mapping1}()$ and $\text{Mapping2}()$ represent the linear mapping operations. These operations are designed to ensure dimensional consistency between the emotional features from the two branches and to prevent excessively high feature dimensionality, thereby reducing the

risk of model overfitting.

4.3.3. Classify and loss function

After the feature extraction module, for the final emotion classification, we combine three fully connected layers and a Softmax activation function to classify emotions of different categories. The operation of the fully connected layers can be represented as follows:

$$s_l = \sigma(f(s_{l-1})) \quad (14)$$

$$f(h) = Wh + b \quad (15)$$

where $f()$ denotes the mapping function performed by the fully connected layer, which linearly converts an m -dimensional vector into an n -dimensional vector by using $W \in \mathbb{R}^{m \times n}$. $b \in \mathbb{R}^n$ indicates the bias, while $\sigma()$ represents the activation function. The first two fully connected layers employ the ReLU activation function, whereas the final fully connected layer utilizes the Softmax activation function. The Softmax function is widely utilized in deep learning models and is especially proficient in handling small sample classification challenges. The classification procedure can be described as follows:

$$p_k = \frac{\exp(w_k^T x)}{\sum_j \exp(w_j^T x)} \quad (16)$$

where p_k represents the probability that sample x belongs to the k -th category of emotions. $\sum_j \exp(w_j^T x)$ denotes the normalization term that ensures that the probabilities sum to 1.

In this study, we employ the cross-entropy loss function to regulate the model. To further prevent overfitting, an L_2 regularization constraint is also applied. The cross-entropy loss is defined as follows:

$$\text{Loss} = - \sum_{n=1}^N \hat{y}_n \log(y_n) + \tau \|\Theta\| \quad (17)$$

where $\hat{y}$ denotes the forward prediction output, y_n denotes the true label, N denotes the number of samples, $\tau \|\Theta\|$ is the regularization constraint term, and τ is the weighting factor.

In this paper, the loss is backpropagated to update the model parameters. The parameters of the neighbor matrix $\mathbf{W}$ of the DGCM can be updated during the gradient backpropagation process. The partial derivatives of $\mathbf{W}$ can be defined as follows:

$$\frac{\partial \text{Loss}}{\partial \mathbf{W}} = \begin{pmatrix} \frac{\partial \text{Loss}}{\partial w'_{11}} & \dots & \frac{\partial \text{Loss}}{\partial w'_{1C}} \\ \vdots & \ddots & \vdots \\ \frac{\partial \text{Loss}}{\partial w'_{C1}} & \dots & \frac{\partial \text{Loss}}{\partial w'_{CC}} \end{pmatrix} \quad (18)$$

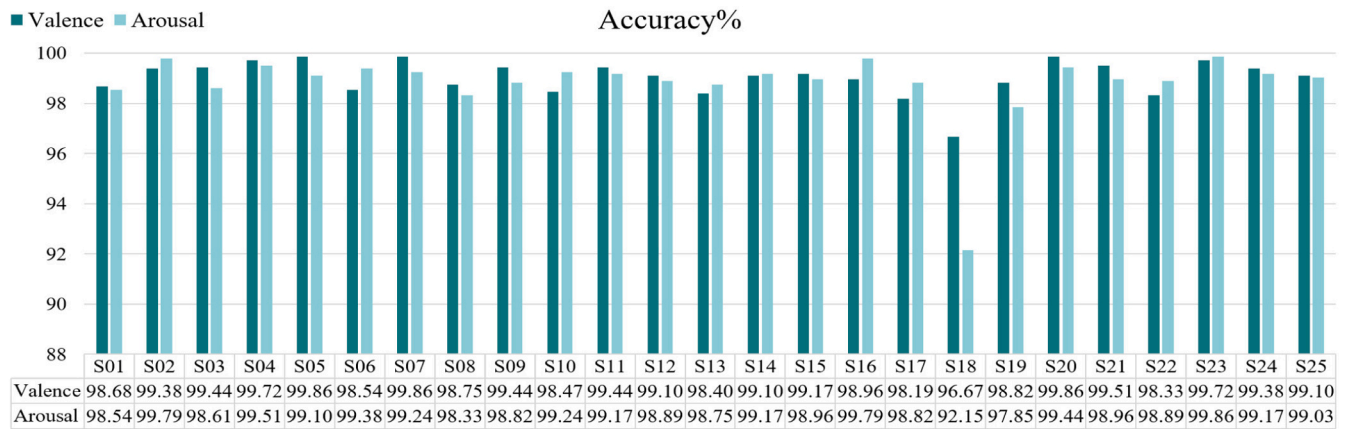
After calculating the partial derivatives of $\frac{\partial \text{Loss}}{\partial \mathbf{W}}$, the optimal adjacency matrix $\mathbf{W}$ is updated by

$$\mathbf{W}' = (1 - \eta)\mathbf{W} + \eta \frac{\partial \text{Loss}}{\partial \mathbf{W}} \quad (19)$$

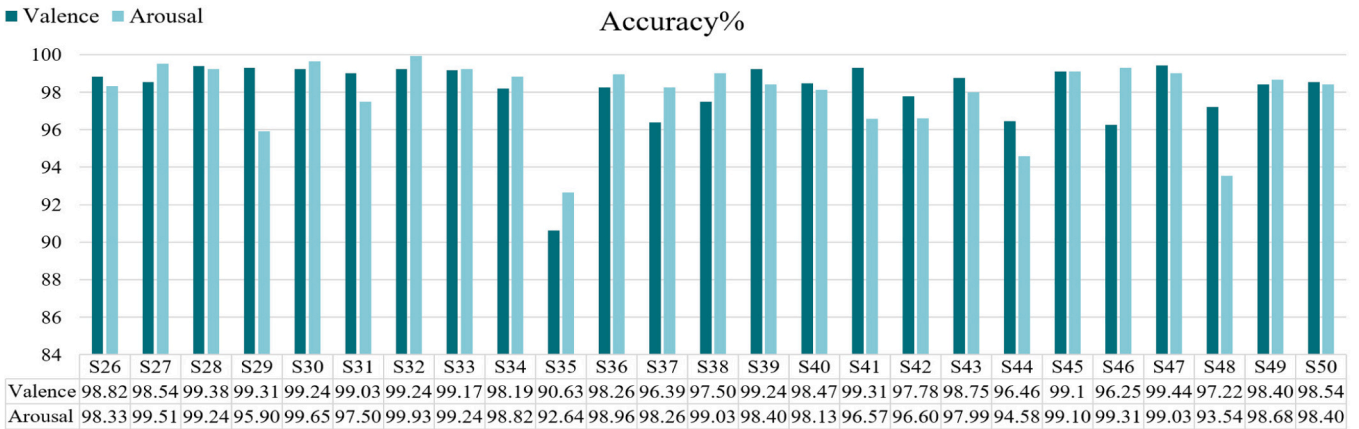
where η is the learning rate of the network.

5. Experimental and result analysis

A number of extensive tests were done to assess the efficacy of the dataset and the proposed method, utilizing both the self-collected HBUE dataset and the publicly available DEAP dataset. This paper first outlines the evaluation metrics employed for emotion recognition algorithms, followed by analyzing the experimental results from various emotion recognition algorithms applied to the two datasets. Additionally, ablation studies were performed to clarify the contributions of each



(a) Test results for subjects 1-25.



(b) Test results for subjects 26-50.

Fig. 8. Accuracy of the proposed method for emotion recognition utilizing the HBUE dataset.

(a) Test results for subjects 1-25.

(b) Test results for subjects 26-50.

module within the proposed model. Finally, brain map visualizations and feature visualizations were carried out to intuitively illustrate the validity of the collected EEG data and the effectiveness of the proposed model.

This paper's hardware environment for validating the algorithm includes an Intel(R) Xeon(R) Silver 4214R CPU operating at 2.40GHz and an NVIDIA GeForce RTX 2080Ti GPU. Experiments were conducted on a Windows 10 operating system using the Pytorch 2.2 framework with Python version 3.11. The learning rate was established at 0.001, the weight decay was set to 0.01, and the model underwent training for 15 epochs. A tenfold cross-validation technique was utilized for model validation. The data for each subject were divided into ten equal parts, with one part used as the test set and the remaining nine parts as the training set. This process was repeated until each part was the test set once. The average of the recognition accuracies obtained across all folds was calculated to represent the final performance for that subject. Subsequently, the average accuracy across all subjects was computed to evaluate the overall emotion recognition performance of the model on the dataset. Furthermore, binary and four-class emotion classification tasks were conducted to validate the constructed dataset and the proposed algorithm comprehensively. The binary classification tasks included high/low valence and high/low arousal. In contrast, the four-class task encompassed the combinations of these dimensions: high valence-high arousal (HVHA), high valence-low arousal (HVLA), low valence-high arousal (LVHA), and low valence-low arousal (LVLA).

5.1. Experiments on the HBUE dataset

In the HBUE dataset, labels for the valence and arousal dimensions were derived via self-assessment, with label values spanning from 1 to 9. In this study, a threshold of 5 was set, where emotion scores less than or equal to 5 were classified as low valence/arousal and scores beyond 5 as high valence/arousal. The binary classification was first performed on the valence and arousal dimensions, with the experimental findings illustrated in Fig. 8. The emotion recognition accuracy of the first 25 participants is shown in Fig. 8(a), while the accuracy of the remaining participants is shown in Fig. 8(b). Experimental results demonstrate that the proposed model achieves average accuracies of 98.57 % and 98.34 % in the binary classification tasks on the HBUE dataset, with overall recognition rates consistently exceeding 90 %. These results highlight the model's effectiveness and stability. Among all participants, Subjects 18 and 35 exhibited the lowest emotion recognition accuracies. This may be attributed to two primary factors. First, from a neuroscience perspective, individual differences in brain neural activity patterns are inherent. In EEG-based emotion recognition, the neural pathways and brain regions associated with valence and arousal may present atypically in Subjects 18 and 35. For instance, brain areas involved in emotional processing—such as the amygdala and prefrontal cortex—may display distinctive characteristics in these individuals, such as reduced neuronal synchronization or weaker electrical signal intensity. These atypical EEG signal patterns can hinder the model's ability to extract features and perform reliable classification accurately. Second, inter-subject variability in physiological characteristics may also

Table 2

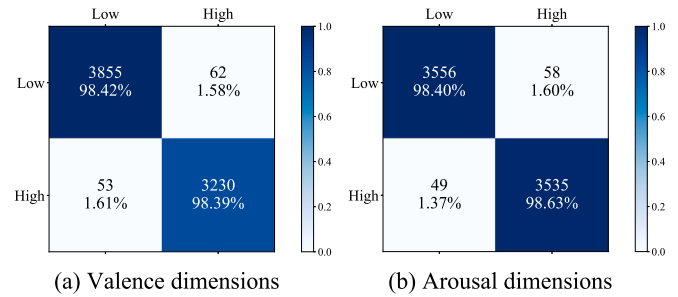
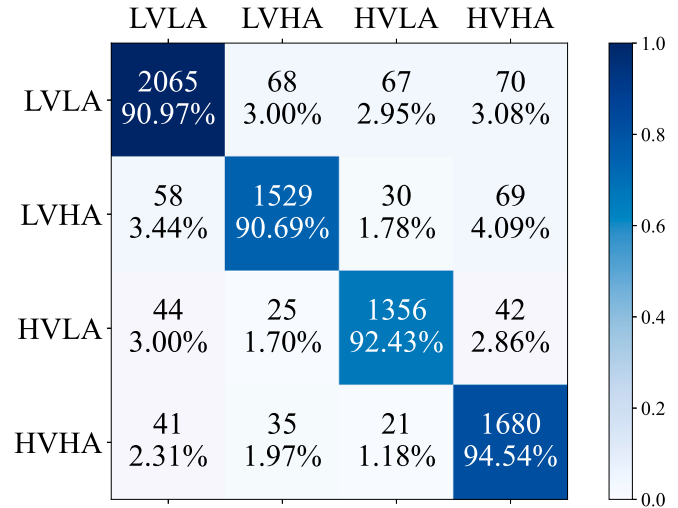
The comparison results of each method on the HBUED dataset.

Method	Accuracy		
	Valence	Arousal	Four-class
KNN (2018)	76.85 %	73.93 %	69.87 %
SVM (2023)	78.11 %	76.57 %	70.20 %
4DCRNN (2020)	96.89 %	96.53 %	90.25 %
DGCNN (2022)	78.34 %	78.95 %	74.19 %
SFCSAN (2022)	94.65 %	94.96 %	88.94 %
AP-CapsNet (2023)	81.45 %	80.91 %	78.38 %
GLFANet (2023)	95.52 %	94.38 %	89.68 %
MAS-DGAT-Net (2024)	96.81 %	96.16 %	90.02 %
IDDA (2024)	97.78 %	96.57 %	91.27 %
LGDAAN-Nets (2025)	97.88 %	97.06 %	91.94 %
Ours	98.57 %	98.34 %	92.15 %

The bold fonts emphasized the best classification accuracy in the tables.

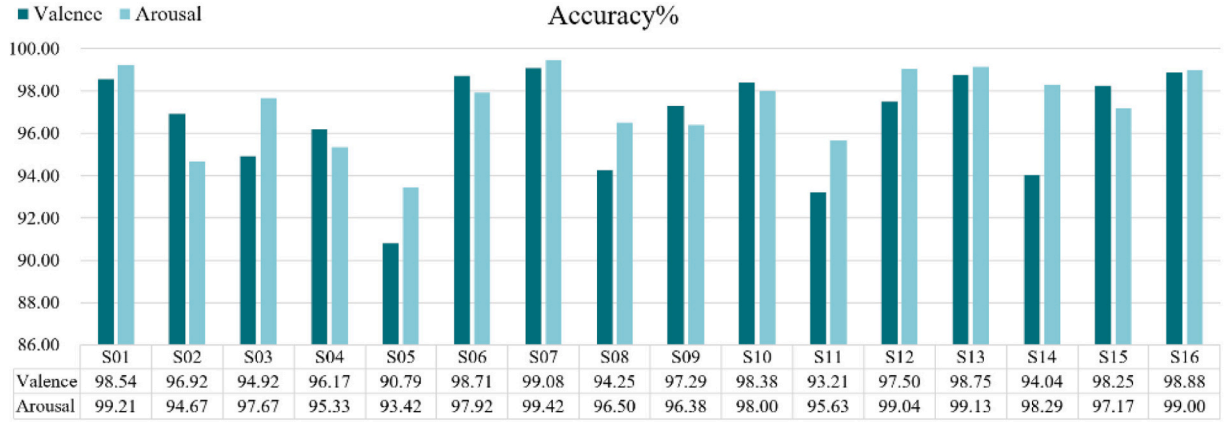
contribute. Subjects 18 and 35 may have relatively thicker scalp tissues or denser hair, which can attenuate or distort EEG signals during transmission. This leads to increased noise in the recorded signals, making feature extraction more challenging and thereby reducing classification accuracy. In addition, differences in subjective emotional experience and external expression further complicate the issue. These two individuals might exhibit less intense or more restrained emotional responses. For example, suppose they possess an emotionally reserved personality. In that case, their physiological responses—such as EEG signal fluctuations—might remain subtle even in states of high arousal or valence, making it more difficult for the model to distinguish and accurately recognize their emotional states. Additionally, it was noted that the model's classification accuracy in both the valence and arousal dimensions showed a strong correlation. Specifically, lower classification performance in the valence dimension was often accompanied by lower results in the arousal dimension. This suggests that individual-specific traits pose a significant challenge in building deep learning models for emotion recognition. When the dataset is small and contains fewer individuals with specific traits, it may lead to inflated classification results. Therefore, constructing large-scale datasets with more participants for model validation is essential.

Several classical machine learning and deep learning methods were selected for comparison to demonstrate the proposed algorithm's advancement. Binary and four-class classification experiments were conducted on the self-constructed dataset to evaluate and compare their performance with that of the proposed method. The specific algorithms used for comparison are as follows: (1) **IDDA** (An et al., 2024): an emotion recognition algorithm based on interconnected dynamic domain adaptation, (2) **SVM** (Tao et al., 2023): an emotion recognition algorithm based on DE features and SVM classifier, (3) **KNN** (Yang et al., 2018): an emotion recognition algorithm based on DE features and the classical KNN classifier, (4) **DGCNN** (Gao et al., 2022): an emotion recognition algorithm based on DE features and DGCNN, (5) **AP-CapsNet** (Liu et al., 2023a, 2023b, 2023c, 2023d): an emotion recognition algorithm using attention mechanisms and pre-trained convolutional capsules, (6) **SFCSAN** (Li et al., 2022): an emotion recognition method based on spatio-temporal convolutional self-attention networks, (7) **4DCRNN** (Shen et al., 2020): an emotion recognition algorithm based on four-dimensional convolutional RNN, (8) **MAS-DGAT-Net** (Liu et al., 2024): an emotion recognition algorithm based on dynamic graph attention networks with multi-branch feature extraction and stage-fusion, (9) **LGDAAN-Nets** (An et al., 2025): an emotion recognition algorithm based on local and global domain adversarial neural networks, (10) **GLFANet** (Liu et al., 2023a, 2023b, 2023c, 2023d): an emotion recognition algorithm based on global to local feature aggregation network. The experimental results are presented in Table 2, with the best classification accuracy emphasized in bold. The results reveal that the proposed model attained superior recognition performance on the HBUED dataset, illustrating the algorithm's efficacy. Moreover, the

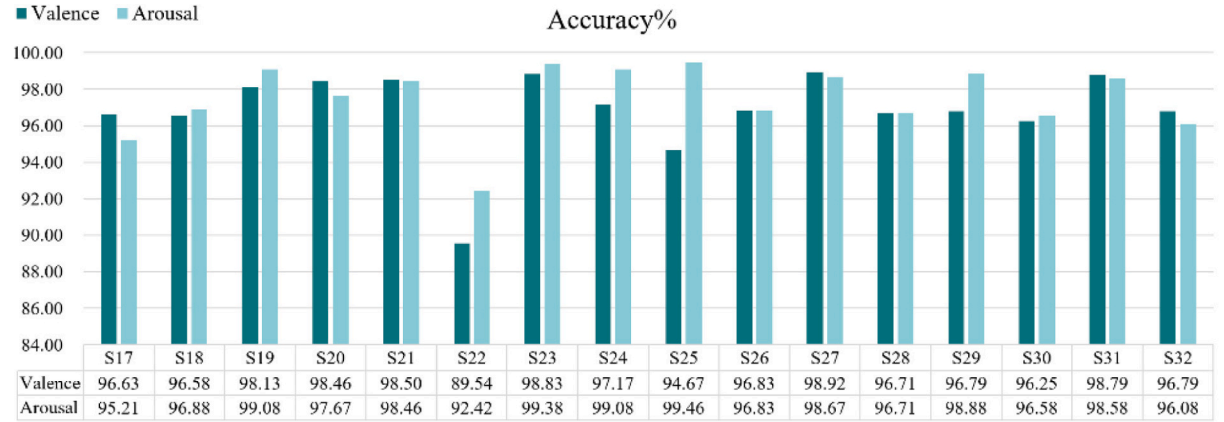
**Fig. 9.** The binary classification confusion matrix on the HBUED dataset.**Fig. 10.** The four-class confusion matrix on the HBUED dataset.

proposed model outperforms DGCNN in terms of accuracy, indicating that the improvements introduced by the TFEM and the PFEM are effective enhancements to the original DGCNN framework. This conclusion will be further validated through ablation studies. In addition, the proposed algorithm achieves the highest classification accuracy in the four-class emotion recognition task, demonstrating its superior feature extraction capability and classification performance.

To visually illustrate the algorithm's performance in recognizing each emotion category, the confusion matrices are presented in Figs. 9 and 10. Figs. 9(a) and 9(b) show the confusion matrices for the valence and arousal dimensions on the HBUED dataset. Fig. 10 displays the confusion matrix for the four-class emotion classification task using the proposed algorithm on the same dataset. In these matrices, the horizontal axis signifies the expected label categories, whilst the vertical axis indicates the actual label categories. Deeper color in the diagonal blocks signifies better algorithm performance. The arousal dimension of the HBUED dataset reveals consistently dark diagonal blocks, signifying a relatively high and balanced recognition accuracy across the arousal dimension. In contrast, for the valence dimension, the matrix block in the upper-left corner is darker, suggesting that the proposed algorithm has a higher accuracy in recognizing negative emotions, making it particularly suitable for early diagnosis of depression. Furthermore, as shown in Fig. 10, the proposed model demonstrates satisfactory performance in the four-class EEG emotion classification task, with most samples correctly classified and recognition accuracies exceeding 90 % for all categories. From a neurological perspective, the high valence category shows higher accuracy in terms of valence, likely because positive emotions are associated with more stable and consistent neural activity patterns that are easier for the model to learn. In contrast, negative emotions involve more complex neural dynamics and greater individual variability, making them more challenging to classify.



(a) Test results for subjects 1-16.



(b) Test results for subjects 17-32.

Fig. 11. Accuracy of the proposed method for emotion recognition utilizing the DEAP dataset.

Regarding the arousal dimension, although both high and low arousal categories achieve relatively high accuracies, physiological variability among individuals can cause signal fluctuations, leading to occasional misclassifications. The HVHA class yields the highest accuracy among the specific emotion categories due to its distinct and stable neural activity patterns. The HVLA class represents a more subdued emotional state, but its features may be confused with those of HVHA. The LVHA class is characterized by more chaotic neural responses and greater individual differences in emotional regulation, contributing to misclassification. Lastly, the LVLA category often exhibits weak and less pronounced emotional signals, making it susceptible to confusion with LVHA.

5.2. Experiments on the DEAP dataset

Analogous experiments were performed to assess the efficacy of the proposed method further utilizing the publicly accessible DEAP dataset. Fig. 11 shows the emotion recognition accuracy for each participant in the DEAP dataset. Among them, Participant 25 achieved the highest accuracy in the arousal dimension at 99.46 %, while Participant 22 had the lowest accuracy at 89.54 %, below the average of 97.40 %. This lower accuracy might be due to this participant's subjective response to the video stimuli not being well captured. The standard deviations for the two dimensions were 2.30 % and 1.79 %, indicating that the proposed model demonstrated relatively steady performance on the DEAP dataset.

In addition, the proposed method was compared with the following baseline algorithms on the DEAP dataset: (1) all the aforementioned comparison methods; (2) KLGEV (Wei et al., 2024): an emotion

Table 3

The comparative results of each model on the DEAP dataset.

Method	Accuracy		
	Valence	Arousal	Four-class
KNN (2018)	77.64 %	78.72 %	72.34 %
SVM (2023)	89.33 %	89.99 %	83.69 %
4DCRNN (2020)	94.22 %	94.58 %	90.16 %
DGCNN (2022)	79.23 %	80.57 %	75.02 %
SFCSAN (2022)	95.15 %	95.76 %	92.87 %
AP-CapsNet (2023)	93.89 %	95.04 %	90.45 %
GLFANet (2023)	94.53 %	94.91 %	92.92 %
MAS-DGAT-Net (2024)	94.69 %	95.24 %	93.46 %
IDDA (2024)	95.22 %	95.43 %	93.28 %
KLGEV (2024)	78.95 %	77.61 %	58.02 %
LGDAAN-Nets (2025)	96.34 %	95.81 %	94.04 %
Hourglass-h (2025)	94.31 %	94.20 %	83.60 %
RLSR (2025)	93.02 %	93.14 %	88.82 %
Ours	96.72 %	97.40 %	94.93 %

The bold fonts emphasized the best classification accuracy in the tables.

recognition algorithm based on spatiotemporal microstate analysis of EEG patterns, (3) Hourglass-h (Liu et al., 2025): an emotion recognition method based on a multi-reservoir echo state network architecture, and (4) RLSR (Gao et al., 2025): a multi-channel EEG emotion recognition algorithm based on low-rank subspace self-representation features. Table 3 presents the performance comparison between these methods and the proposed approach on the DEAP dataset. Experimental results show that the proposed algorithm achieves the best emotion recognition performance in both valence-arousal binary classification and four-class emotion classification tasks, further validating the effectiveness of the

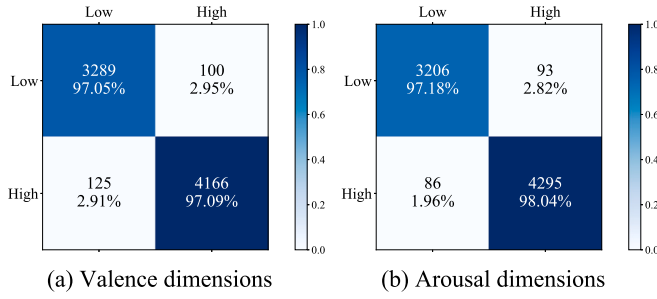


Fig. 12. The binary classification confusion matrix on the DEAP dataset.

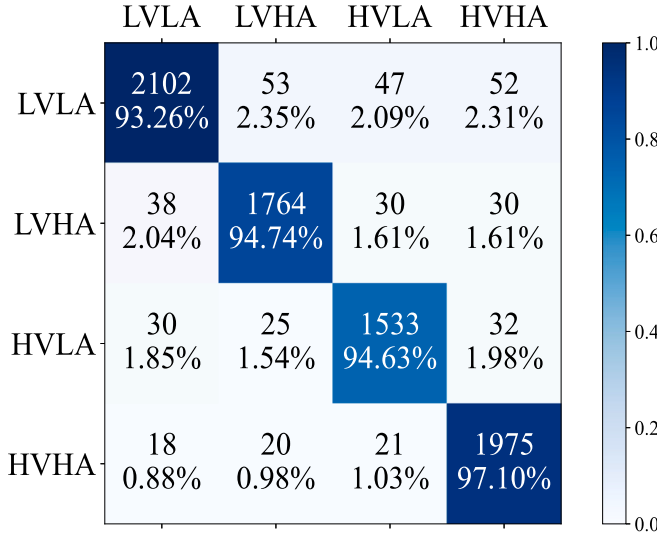


Fig. 13. The four-class confusion matrix on the DEAP dataset.

proposed method.

The confusion matrix presented in Fig. 12 demonstrates that the proposed algorithm is more proficient in identifying positive emotions

than negative emotions within the DEAP dataset. This indicates that the proposed method may serve as a beneficial supplementary diagnostic instrument for preventing psychological problems. As shown in Fig. 13, the proposed model demonstrates strong performance in the

four-class EEG emotion classification task, achieving recognition accuracies above 93 % across all categories. From a neurological perspective, high valence categories (HVLA and HVHA) exhibit higher accuracies in the valence dimension. This is because positive emotional states are typically associated with more stable patterns of neural activity in relevant brain regions, resulting in the related EEG features that are easier for the model to learn and capture. In contrast, low-valence categories (LVLA and LVHA) involve more complex neural activity influenced by individual experiences, personality traits, and other factors. This leads to greater inter-subject variability and makes emotion recognition more challenging. In the arousal dimension, high arousal (LVHA, HVHA) and low arousal (LVLA, HVLA) categories achieve relatively high accuracy overall. However, variations in individual physiological characteristics and neural activity can introduce fluctuations in the EEG signals, causing occasional misclassifications. At the category level, the HVHA class achieves the highest accuracy at 97.10 %, as the strong and positive emotional state produces distinct and stable neural activity patterns, facilitating accurate model recognition. The HVLA class achieves an accuracy of 94.63 %; while still in a positive emotional state, it is more subdued, and its EEG features partially overlap with those of HVHA, leading to misclassifications. The LVHA class achieves an accuracy of 94.74 %; due to the chaotic neural activity associated with negative and high arousal emotions and individual differences in emotional regulation mechanisms, the model may struggle to classify this category accurately. Lastly, the LVLA class achieves the lowest accuracy among the four, at 93.26 %, as this weak and low arousal negative emotional state results in subtle EEG signal variations that can be easily confused with those of LVHA, leading to occasional classification errors.

5.3. Friedman test and subsequent Nemenyi test

To perform a more comprehensive comparison of the emotion recognition algorithms, we employ the Friedman test followed by the Nemenyi test to demonstrate the advantages of the proposed method.

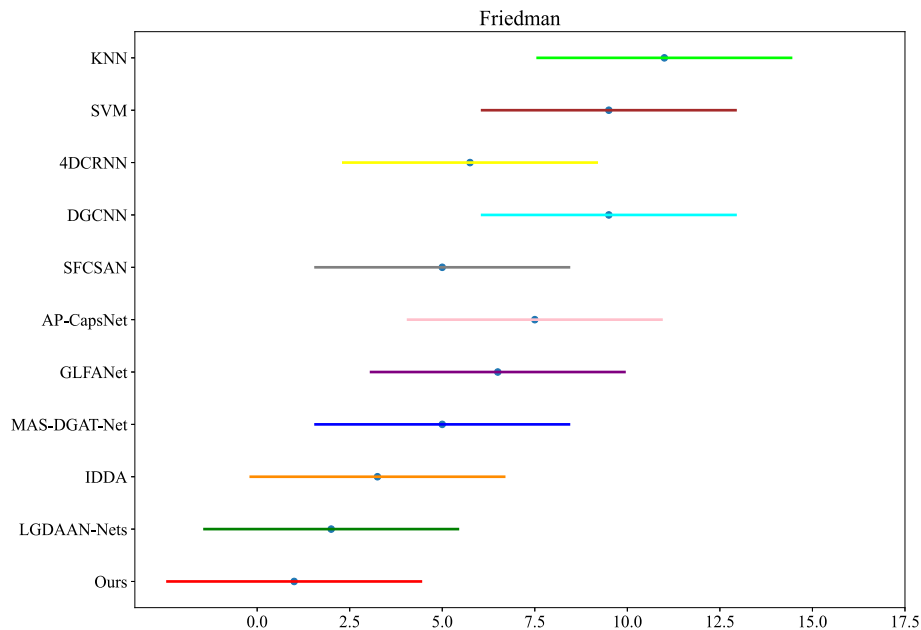


Fig. 14. Friedman hypothesis testing plot. The dots are the mean values of each algorithm for different datasets and dimensions if the absence of overlap between the two algorithms indicates that they are significantly dissimilar.

Table 4
Results of ablation experiments.

Model	Accuracy	
	Valence	Arousal
w-TFEM	80.93 %	80.10 %
w-PFEM	90.21 %	92.70 %
w-GFEM	94.00 %	93.72 %
Ours	98.57 %	98.34 %

The Friedman test is a non-parametric statistical test that does not depend on assumptions about data distribution, making it applicable when data do not meet the requirements of normal distribution and homoscedasticity. The basic hypothesis of the Friedman test is that all algorithms perform equally. In the experiments, comparing the proposed algorithm with the baseline methods, the calculated indicator τ_F was 80.9999, higher than $\alpha = 0.05$, and the Friedman test statistic was 2.448. As a result, the hypothesis that “all algorithms perform equally” was rejected. Subsequently, the Nemenyi test was conducted. Under the condition of $q_{0.05} = 2.949$, the critical value range $CD = 4.5047$ was obtained from the Nemenyi test. Based on the critical value range and the Friedman test, Fig. 14 was plotted. The vertical axis in Fig. 14 represents the various algorithms, while the horizontal axis shows the mean rank. A dot represents each algorithm’s mean rank, and a horizontal line centered on the dot indicates the critical value range. If the horizontal lines of two algorithms overlap, it suggests no significant difference between them; otherwise, a significant difference exists. As illustrated in Fig. 14, the proposed algorithm has the highest mean rank, indicating it performs the best. A significant difference exists between the proposed algorithm and DGCNN, KNN, and SVM.

5.4. Ablation experiments

This study conducted an ablation analysis utilizing the self-collected HBUE dataset to elucidate the function of each module in the proposed algorithm. The models constructed for the ablation study were as follows: (1) w-TFEM: retaining only the TFEM, making the model single-input. (2) w-PFEM: retaining only the PFEM and making the model single-input. (3) w-GFEM: removing the IFEM based on the attention mechanism while keeping all other structures intact. Table 4 shows the experimental results.

Table 4 indicates that the proposed model surpasses other structures in emotion recognition. The w-TFEM model maintains the TFEM while eliminating the PFEM, resulting in a single input structure. The accuracy of this model decreases by 17.64 % and 18.24 % compared to the proposed model, representing the most significant drop among all tested structures. This suggests that the PFEM significantly influences model performance. The w-PFEM model retains the PFEM while eliminating the TFEM, resulting in a single input structure. The performance decline in this configuration suggests that the TFEM contributes crucial spatial topology information, making the fused feature information more comprehensive. Ultimately, the w-GFEM model exhibits a reduction in recognition accuracy relative to the proposed model. This decline signifies that the IFEM, grounded in the attention mechanism, improves the efficacy of the TFEM by seizing critical channel information pertinent to emotions. This, in turn, facilitates the subsequent w-DGCM in examining inter-channel correlations.

5.5. Brain map visualization

To further investigate the accuracy of neural activation patterns in the self-collected HBUE dataset, in this study, we selected one subject

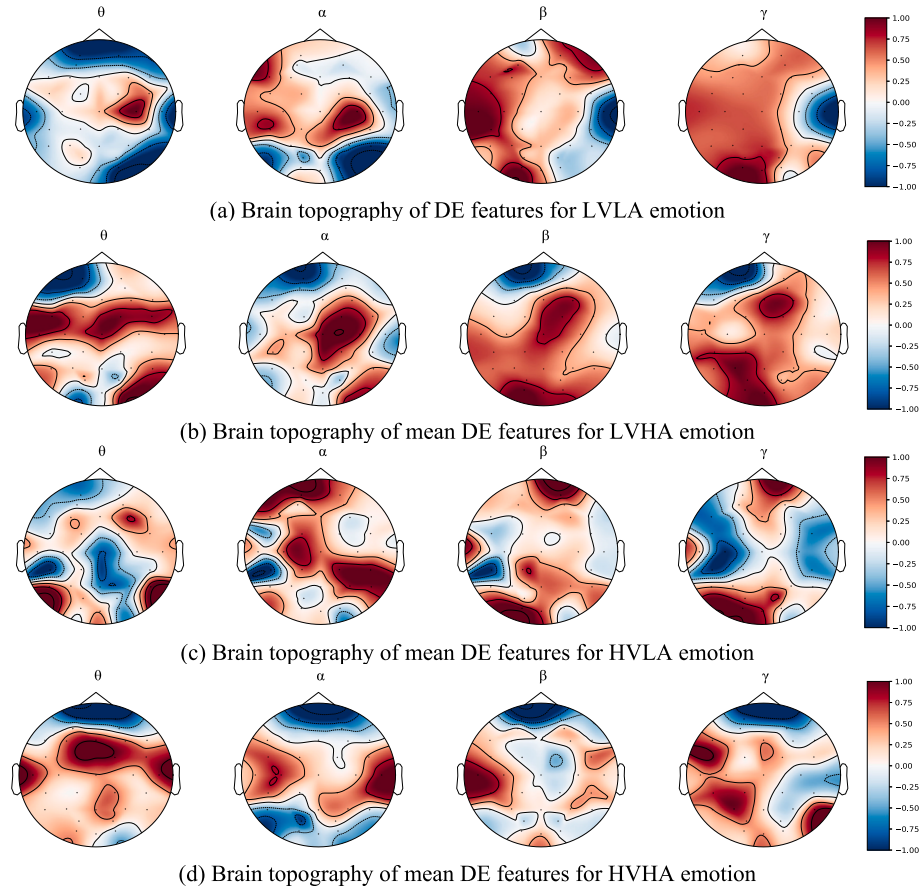


Fig. 15. Brain topographies of individual emotional mean DE features for Participant 5 on the HBUE dataset.

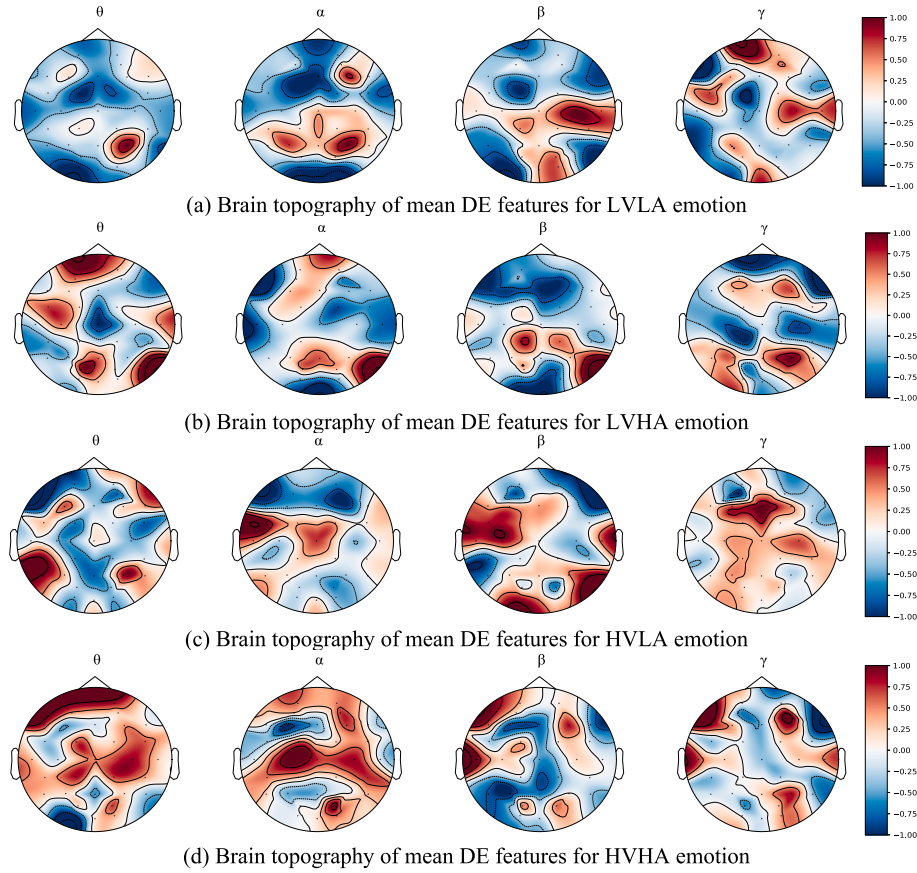


Fig. 16. Brain topography of individual emotion mean DE features for Participant 1 on the DEAP dataset.

from the HBUED and DEAP datasets to illustrate the mean DE features throughout five frequency bands for different emotions. Fig. 15 shows the brain topography of Participant 5 from the HBUED dataset, while Fig. 16 displays the brain topography of Participant 1 from the DEAP dataset. Figs. 15(a) and 16(a) illustrate the average DE features under LVLA emotions. Fig. 15(b) and Fig. 16(b) depict LVHA emotions. Figs. 15(c) and 16(c) show HVLA emotions. Figs. 15(d) and 16(d) represent HVHA emotions. The four columns in each figure correspond to frequency bands θ , α , β , and γ , respectively. From Figs. 15 and 16, the following conclusions can be drawn: (1) The neural activation associated with low valence/low arousal emotions is generally lower than that of high arousal/high valence emotions, consistent with the findings in the conclusions by Peng et al., 2023. In the HBUED dataset, this phenomenon is most pronounced in the θ frequency band. (2) EEG channels abundant in emotional data are predominantly situated in the frontal and temporal lobes, consistent with the results in (Lin et al., 2010; Zheng and Lu, 2015; Wu et al., 2020). (3) The level of activation across different brain regions may vary between datasets. For example, the activation levels across various frequency bands in the HBUED dataset tend to be higher, whereas in the DEAP dataset, they are generally lower. This observation is consistent with the conclusions of (Li et al., 2021) and is likely due to differences in the participants' age, knowledge structure, and the type of emotional stimuli used.

5.6. Feature visualization

To visually demonstrate the classification efficacy of the proposed model, we randomly picked one participant from both the self-compiled HBUED dataset and the publicly accessible DEAP dataset. Subsequently, t-distributed stochastic neighbor embedding (t-SNE) was utilized to show the distributions of the raw data and the model outputs (Van Der Maaten and Hinton, 2008). Fig. 17 illustrates the visualization results for

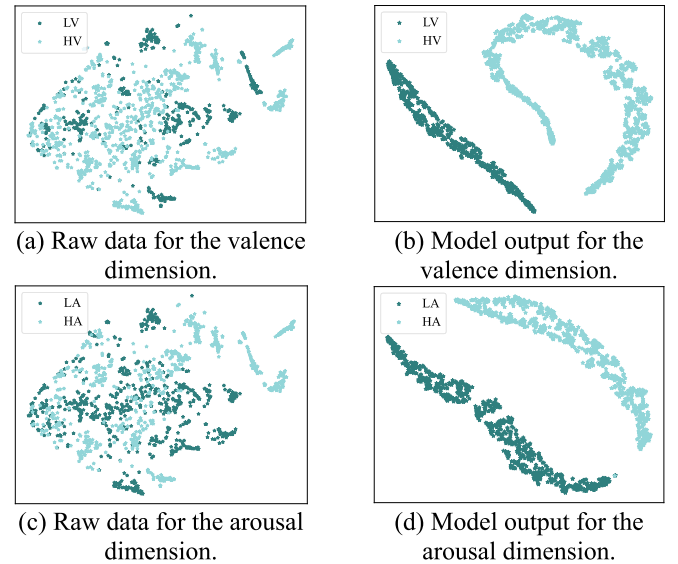


Fig. 17. Visualization of the features of subject 5 in the HBUED dataset.

the HBUED dataset, whereas Fig. 18 depicts the results for the DEAP dataset. Figs. 17(a) and 18(a) depict the original data distribution for the valence dimension. Figs. 17(b) and 18(b) show the model output distribution for the valence dimension. Figs. 17(c) and 18(c) depict the original data distribution for the arousal dimension. Fig. 17(d) and Fig. 18(d) depict the model output distribution for the arousal dimension. In these visualizations, LV indicates low valence, HV indicates high valence, LA indicates low arousal, and HA indicates high arousal. As

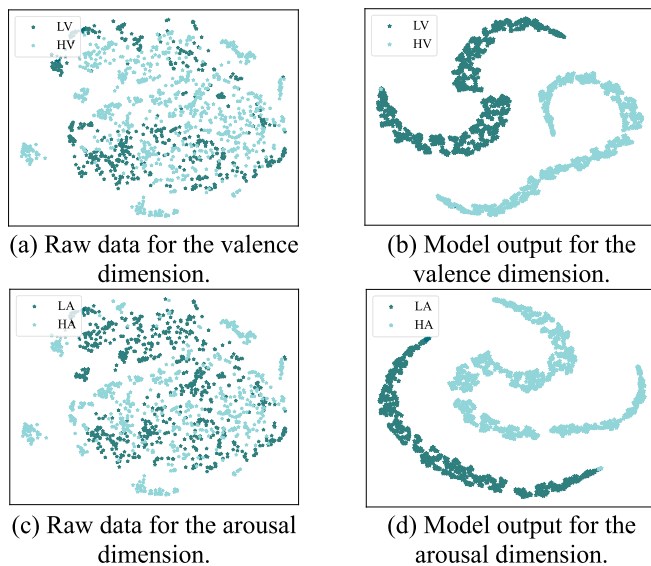


Fig. 18. Visualization of the features of subject 1 in the DEAP dataset.

shown in Figs. 17 and 18, the original EEG data distributions are chaotic, with minimal differences between the two classes, making them hard to distinguish. After processing with the proposed model, the class differences in the output data increase significantly, facilitating easier differentiation. The visualization results demonstrate that the proposed model achieves good classification performance on both the HBUE and DEAP datasets.

6. Conclusion

Datasets are a crucial foundation for scientific research. To address the issue of limited participant numbers in existing EEG emotion datasets, we have collected an EEG emotion dataset comprising 50 participants. This work provides valuable materials for researchers and will further advance BCI technology. A deep learning algorithm for EEG emotion recognition has also been proposed, featuring a dual-input structure with complementary information between the two branches. Ablation studies have demonstrated the significance of each module in this algorithm. However, the participants in the dataset come from a single source (students from one University), which may overlook the effects of factors such as gender and culture on the experimental outcomes. Future work will focus on diversifying participant sources and analyzing the impact of these factors on the experiments.

CRedit authorship contribution statement

Shuaiqi Liu: Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Conceptualization. **Xinrui Wang:** Writing – review & editing, Writing – original draft, Methodology. **Yanling An:** Writing – original draft, Software, Methodology, Conceptualization. **Zeyao Wang:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation. **Zhihui Gu:** Writing – review & editing, Formal analysis, Data curation. **Yudong Zhang:** Writing – review & editing, Formal analysis. **Shuhuan Zhao:** Writing – review & editing, Project administration, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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